Keegan Lemos

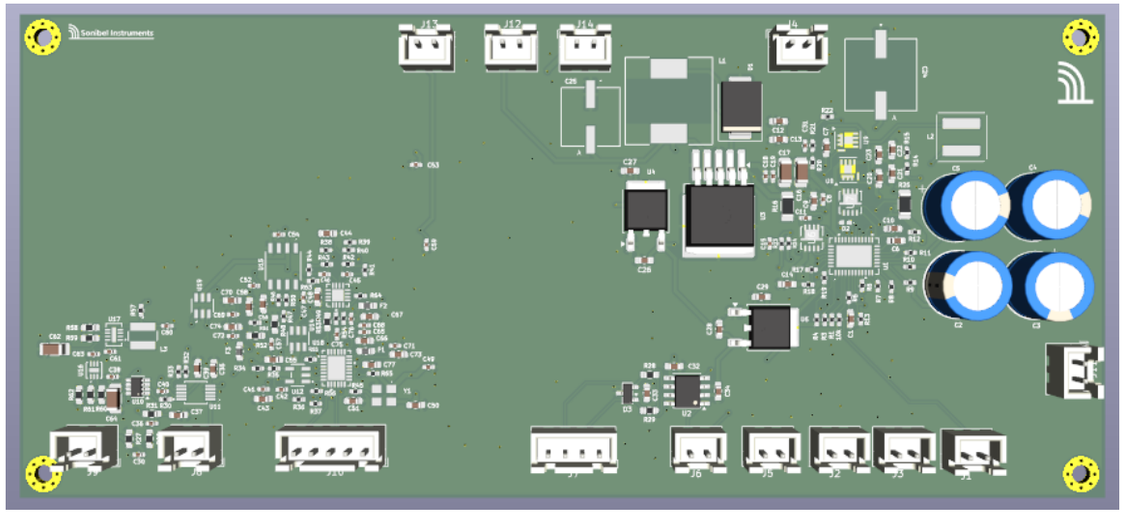
Electrical Engineering, University of British Columbia

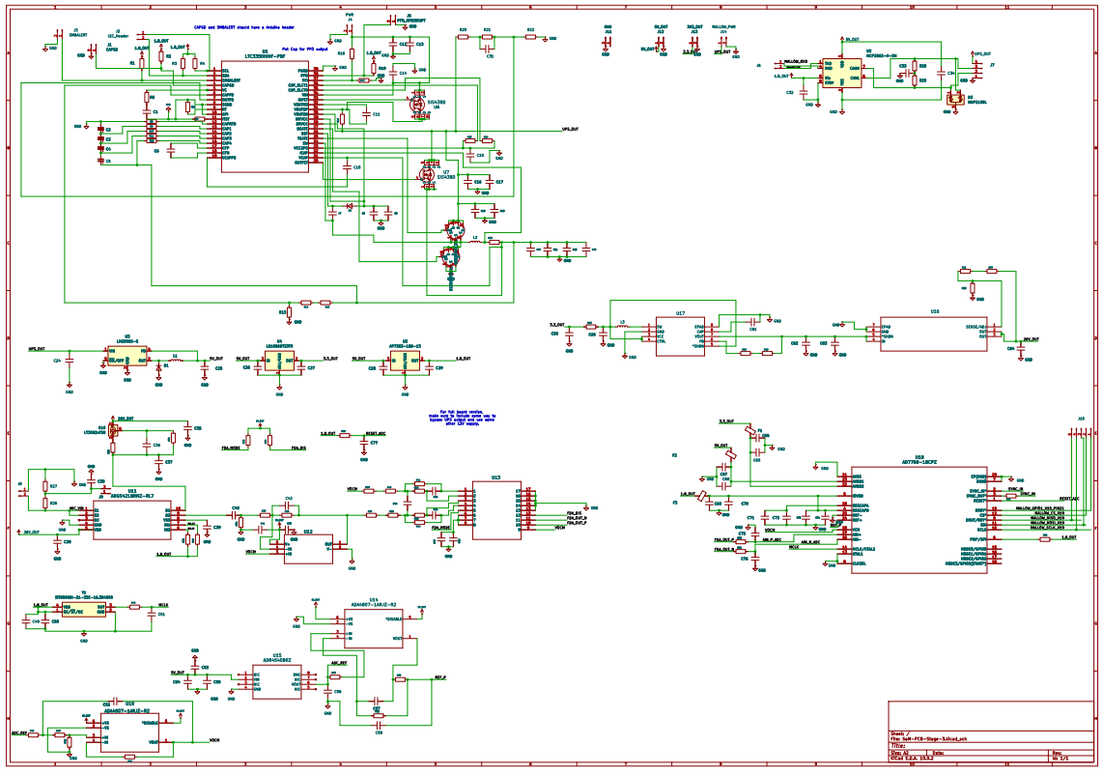
k.lemos0504@gmail.com

linkedin.com/in/keegan-lemos

Weld data acquisition board

SoM - Stage 3 / 6 layers / KiCad





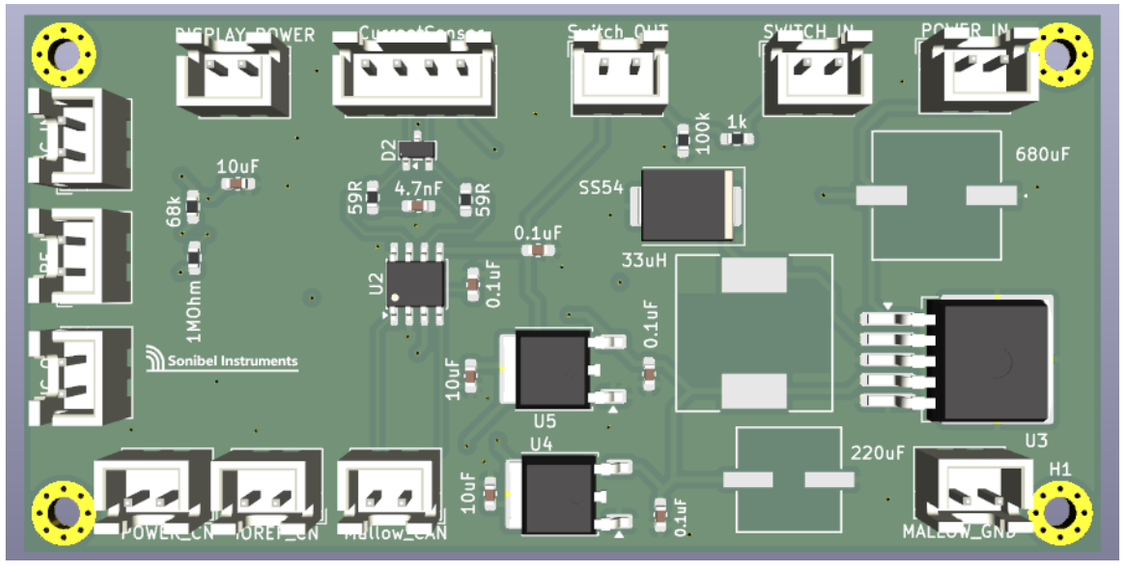
Schematic / SoM-PCB-Stage-3.pdf

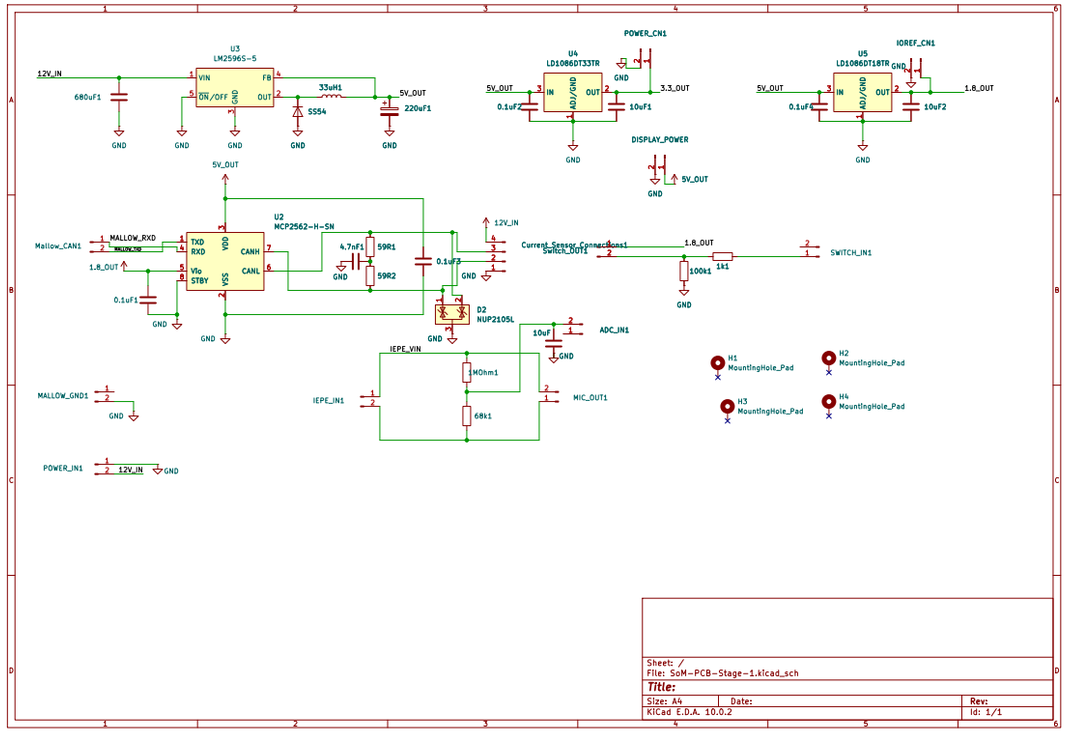
I designed and routed this 6-layer board to record weld audio and connect to the current sensor over CAN. It brings the IEPE microphone input, 24-bit ADC, and supercapacitor backup supply onto one board. The ADC sends samples to the system-on-module over SPI, while the board supplies the 5 V, 3.3 V, and 1.8 V rails.

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Peripheral power & interfaces

SoM - Stage 1 / KiCad





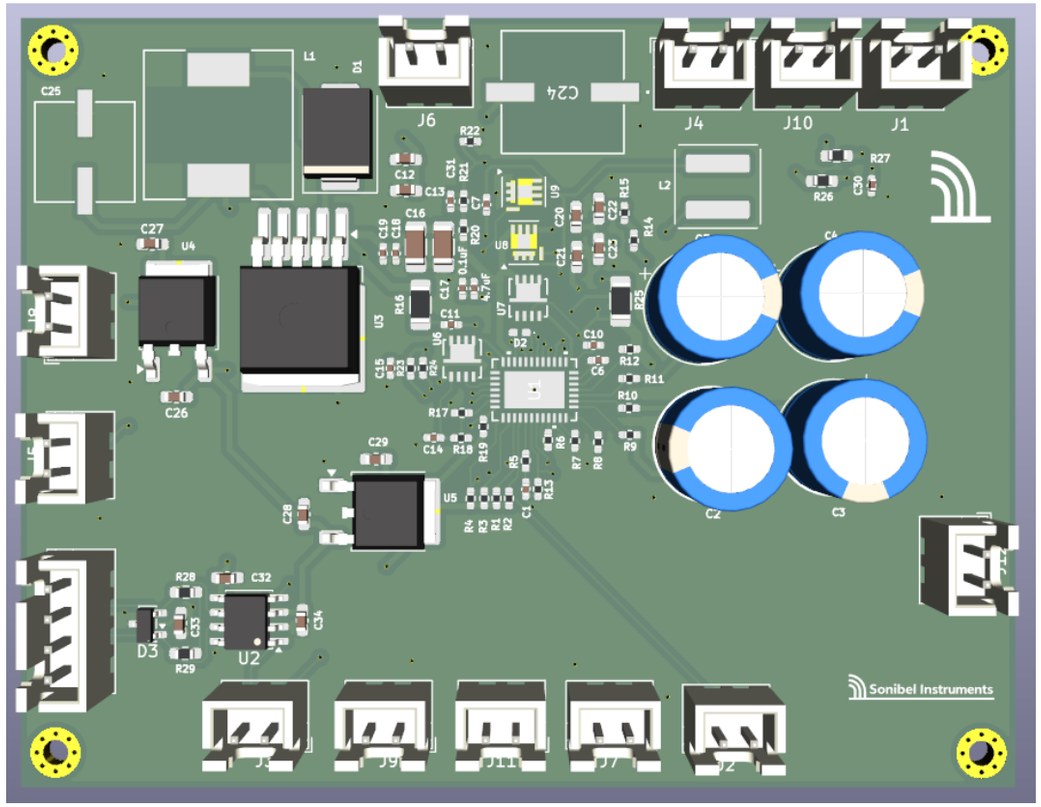
Schematic / SoM-PCB-Stage-1.pdf

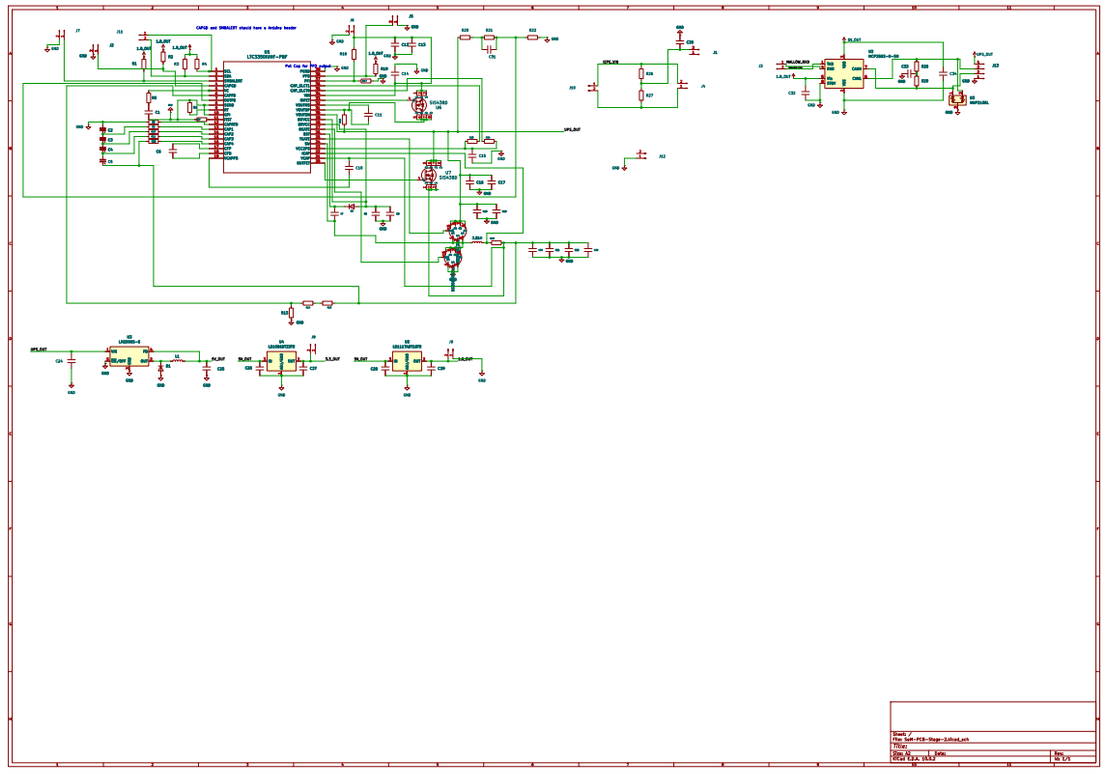
I designed this first version to connect sensors, a display, and switches to the system-on-module. It converts a 12 V input into 5 V, 3.3 V, and 1.8 V supplies and includes CAN and microphone connections. The CAN interface has termination and transient protection, and the microphone path is AC-coupled.

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Adding backup power

SoM - Stage 2 / KiCad





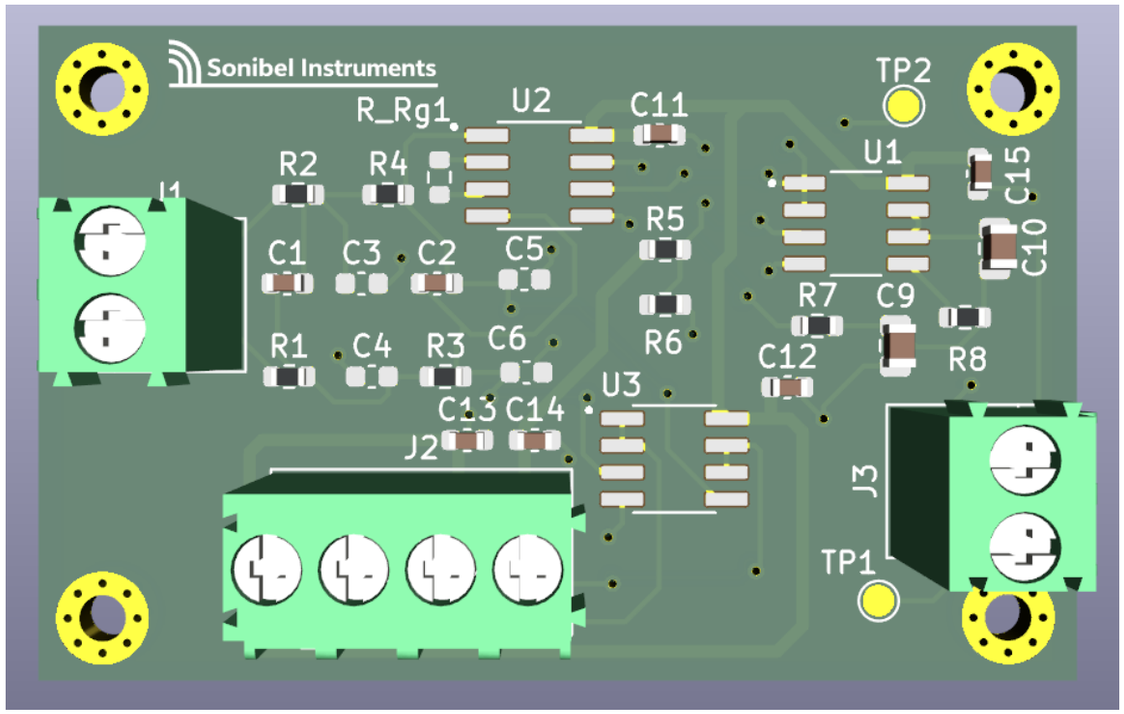
Schematic / SoM-PCB-Stage-2.pdf

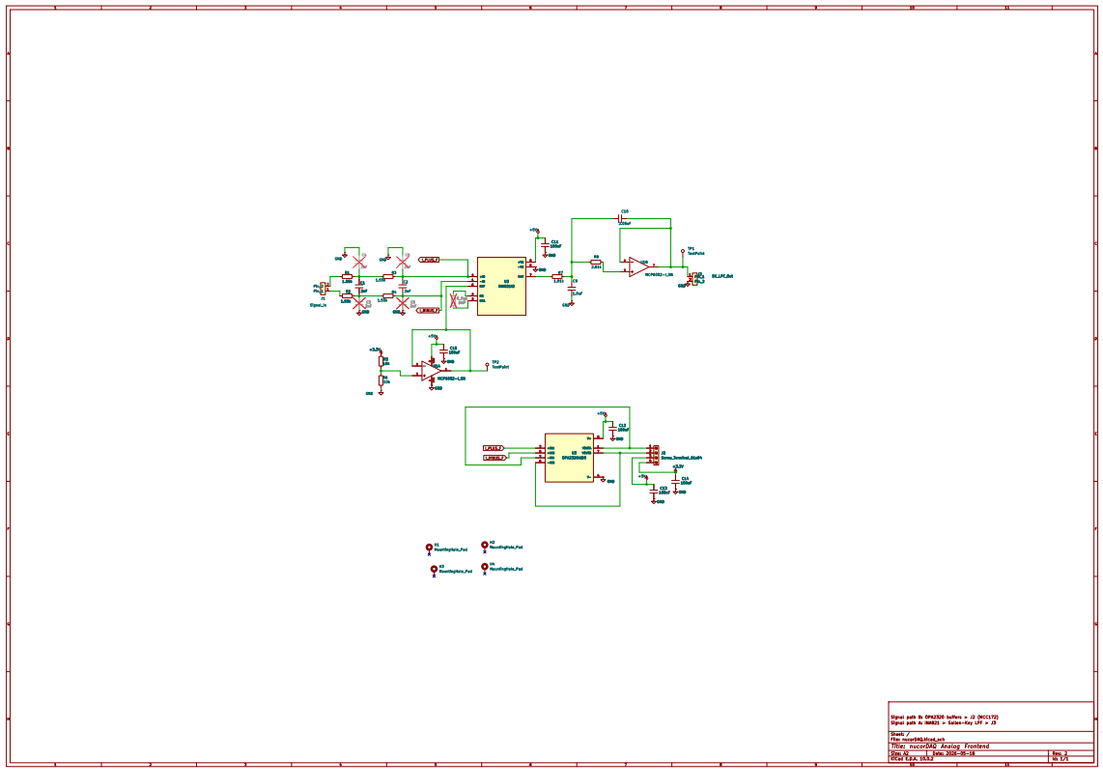
For the second version, I added an LTC3350 backup supply with four supercapacitors. This gives the system time to save its recordings and shut down when input power is lost. I also brought out power-failure and capacitor-status signals so the host can monitor the supply, while keeping the earlier board's voltage rails and CAN interface.

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Current-sensor analog front end

nucorDAQ / 4 layers / KiCad





Schematic / nucorDAQ.pdf

I designed and routed this 4-layer board to condition a current-sensor signal before recording it. One path uses an INA821 instrumentation amplifier followed by a 50 Hz Sallen-Key low-pass filter; the other uses OPA2320 buffers to feed an MCC172 acquisition module. I included a buffered reference and test points for checking the signals during bring-up.